Introduction to Polymers as an Engineering Material

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Lecture - 11

Introduction to Elastomers and its applications

What are elastomers

As per ASTM, an elastomer is defined as “a polymeric material, which, at room temperature, can be stretched to at least twice its original length and upon immediate release of stress, will return quickly to approximately its original length”.

The term derives from “elastic”-one of the fundamental properties of rubber

Elastomers share many characteristics with plastics:

1. They are essentially non-crystalline in structure.
2. They are nonconductors of electricity and are relatively low heat conductors.
3. They are high in resistance to chemical and corrosive environments.
4. They have relatively low softening temperatures.
5. They exhibit viscoelastic behavior generally to a greater extent than plastics.
6. They oxidize or age, causing deterioration and changes in properties, generally more than so than plastics

ELASTOMERS VS PLASTICS

Elastomers	Plastics
Hydrocarbons	Hydrocarbons
Completely amorphous but may crystallize Upon stretching	Amorphous to crystalline
Extensibility: 100-1000% At RT	Extensibility: generally less than 50% at RT. Some grades of plastics approach rubber like STATE, e.G. Certain polyethylene's. Some Olefins, styrene's, fluoroplastics and Silicones have elastomer grades
High energy storing capacity even after they strained several hundred percent, vertually complete recovery is achieved once the stress is removed	They do not possess this property

Elastomers	Plastics
The structure consists of very long and geometrically irregular molecular Chains in highly coiled helix form and get Uncoiled upon loading.	The molecular chains are relatively short and Much less coiled.
Generally cross linked after vulcanization. In the unvulcanized state, molecular chains are Linear and only weakly joined to each other by secondary bonds.	Thermoplastics are generally not cross Linked
Thermoset elastomers are not meltable	Thermoplastics are Meltable
Have lower elastic modulus and strength	Have higher elastic mod. And strength

Any linear polymer can be made good elastomer if it satisfies the following requirements:

1. The polymer chain must be very long and geometrically irregular so that thermal agitation will result in strong entanglement and coiled type of arrangement.

2. Intermolecular forces between molecular chains at R.T. thermal energy should be just enough to maintain them in constant mobility.

3. They must have a C=C double bond along the molecular chain to explore the possibility of cross linking which can be introduced in controlled amount to enhance rigidity of the elastomers.

The basic building block of rubber is diene containing C=C double bond as shown below

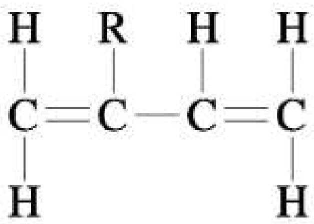

Fig. 1 Dine Monomer

Natural rubber is a lightly cross-linked material. It is polyisoprene. The structure of natural rubber (polyisoprene) is referred to as cis-1,4 polyisoprene where all the unsaturated bonds lie on the same side of the molecule in the 1,4 positions:

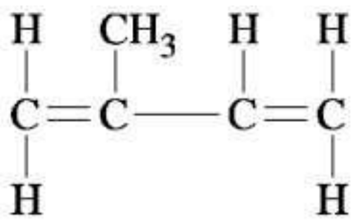

Fig. 2 Isoprene Monomer

Natural Rubber

It is derived from natural rubber latex obtained from a tree called “Hevea Brasiliensis”. The latex is a white milky liquid containing suspension of fine particles of rubber. The latex from the trees is collected, coagulated with formic acid and washed with water. The material is then compressed between the pair of rolls to remove water and produce a sheet material. The sheets are then dried either in hot water or in smokehouses at about 50°C.

Properties of Natural rubber

1. Soft and sticky under the action of heat.
2. Flexible and has large deformability.
3. Low modulus coupled with large elongations make the material resilient i.e. ability to absorb shocks.
4. Tensile strength is low.
5. Creep resistance is low.
6. Poor toughness.
7. Water repellent
8. It shows resistant to alkalises and weak acids

Properties that are improved by vulcanization of natural rubber

1. Great flexibility
2. Excellent wear and abrasion resistance
3. Stiffness and tensile strength
4. Good chemical resistance
5. Less permeable to gases.

Oil-resistant elastomers

1. Buna-N, a copolymer of butadiene and acrylonitrile and also known as Nitrile butadiene rubber (NBR) is developed to improve oil resistance of polybutadiene.
2. It is more expensive than butadiene rubber or SBR and is therefore find limited applications where oil resistance is required.
3. NBR also has improved resistance to degradation from oxidation and finds some applications whre that property is important.
4. NBR has good abrasion resistance but poor low temperature elasticity.
5. Typical applications of NBR include oil & fuel lines, gaskets, seals, conveyor belts and coatings.

Buna-N/Nitrile/NBR

Main Properties:

1. Max Temperature Range from ~ -54°C to 121°C (-65° – 250°F).
2. Very good resistance to oils, solvents and fuels.
3. Good abrasion resistance, cold flow, tear resistance.
4. Preferred for applications with Nitrogen or Helium.
5. Poor resistance to UV, ozone, and weathering.
6. Poor resistance to ketones and chlorinated hydrocarbons.

Most Often Utilized in:

Aerospace & Automotive Fuel Handling Applications

Neoprene

The neoprene family of synthetic rubbers is produced by the polymerization of chloroprene and is also known as polychloroprene or Chloroprene (CR).

Properties and application of Neoprene

1. Chloroprene is another oil-resistant elastomer. It is also termed as Neoprene.
2. Presence of chlorine atom in the monomer improves oil resistance and because it is somewhat bulky, it improves intermolecular interference, strength and stiffness over unmodified polybutadiene, polyisoprene and natural rubber.
3. Thermal stability of neoprene is also better than in polybutadiene or polyisoprene.
4. The presence of chlorine makes neoprene nonflammable, hence applications requiring this property and elasticity are common. Mattresses used in Navy, is a typical example requiring these properties. The mattresses must be nonflammable because of the highly dangerous consequences of a fire on ships, yet the resiliency of the elastomer results in good comfort.
5. Other applications of neoprene are fuel hoses, boots, shoe soles and coatings for fabrics where oil resistance and perhaps, nonflammability are important.


THANK
YOU